Jitendra Singh

Curriculum Vitae

IIT Mandi South Campus
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Career Objective

A dedicated researcher in Theoretical Computer Science and autonomous systems with a focus on algorithmic randomness, algorithmic information theory, and infinite sequences. Eager to expand my knowledge, collaborate with leading experts, and contribute to cutting-edge research.

Education

Aug 2024–Present **Ph.D. in Theoretical Computer Science**, *IIT Mandi*
Focus: Algorithmic Randomness, Algorithmic Information Theory, Infinite Sequences.

2013–2015 **M.Tech in Computer Science (Gold Medalist)**, *MMM University of Technology, Gorakhpur*
Percentage: 78.4%

2008–2012 **B.Tech in Computer Science**, *GBTU / UPTU, Lucknow*
Percentage: 68.75%

Work Experience

Nov 2022–Jul 2024 **Assistant Professor**, *Sharda University, Greater Noida*
Taught courses on computer networks, cryptography, and machine learning. Participated in academic committees (NBA and NAAC); mentored students; coordinated admission teams and contributed to departmental development.

Research Experience and Projects

GPS-Denied Drone Navigation for Martian Terrain – IRoC-U 2025 (ISRO) Principal Coordinator. Guided a student team to the **Second Place** in IRoC-U 2025. Designed and implemented a novel GPS-denied **random navigation algorithm** for an autonomous drone to traverse rough Martian-like terrain and identify a safe landing spot. Successfully completed the mission with high precision during field trials at the ISRO Centre, Bengaluru.
Key Technologies: SLAM, sensor fusion, probabilistic navigation, autonomous landing, path planning under uncertainty.

Autonomous Lunar Exploration Rover – IRoC-U 2024 (ISRO) Principal Coordinator. Developed autonomous path planning and control algorithms for a lunar rover with sample pickup/drop-off capabilities. Integrated modifications to A* search and optimized control algorithms for complex terrain navigation. Budget: Rs 2.9 Lakh. **Awarded by the Honourable President of India** for excellence in innovation.
Key Technologies: A* search, obstacle avoidance, terrain analysis, control systems, autonomous sample handling.

Brain Controlled Prosthetic Limbs (GoI Funded and incubated at IIT Mandi): **Co-Founder**. Co-founded a startup developing a lightweight, carbon-fiber prosthetic arm controlled via non-invasive brain signals. Designed and integrated fine-grained control using EMG sensors (Elemyo) and servo-actuated

mechanisms, enabling natural finger movements through fishing-line tendon systems. Led the system design, hardware–software integration, and testing for improved dexterity, responsiveness, and user comfort.
Key Technologies: EMG signal processing, brain–computer interface (BCI), servo control systems, carbon-fiber fabrication, prosthetic biomechanics.

Research Publications

- Jitendra Singh, Vimal Kumar, Rakesh Kumar, "An Efficient and Secure RSA-Based Certificate-less Signature Scheme for Wireless Sensor Networks." SIRS 2015, Springer Advances in Intelligent Systems and Computing (2016). (SCOPUS)
- Jitendra Singh, Rakesh Kumar, Ajai Kumar Mishra, "Clustering Algorithms for Wireless Sensor Networks: A Review," IEEE ICCSD (2015). (SCOPUS)
- Jitendra Singh, Rakesh Kumar, Vimal Kumar, Ajai Mishra, "Intruder Detection by Visual Cryptography in Wireless Sensor Networks," IEEE IC-CSP (2015). (SCOPUS)
- Ajai Mishra, Jitendra Singh, Rakesh Kumar, "A Review on Fuzzy Logic-Based Clustering Algorithms in WSNs," IEEE (2015). (SCOPUS)
- Ajai Mishra, Rakesh Kumar and Jitendra Singh, "A grid-based approach to prolong lifetime of WSNs using fuzzy logic," Advances in Computational Intelligence (2017). (SCOPUS)

Skills

Theoretical	Algorithmic Randomness, Algorithmic Information Theory, Infinite Sequences
Algorithms	Path Planning, Control Algorithms, A* Search
Mathematics	Linear Algebra, Discrete Probability, Cryptography
Other	Cloud Deployment, Wireless Sensor Networks, Autonomous Systems

Achievements

- Gold Medalist in M.Tech (CSE).
- Awarded by the Honourable President of India for excellence in the Autonomous Lunar Exploration Rover-IRoC-U project.
- GATE Qualified (2013, 2014).
- UGC-NET Qualified (2022–23).
- Paper Reviewer: ICCCIS 2023.